

Q.NO: 6

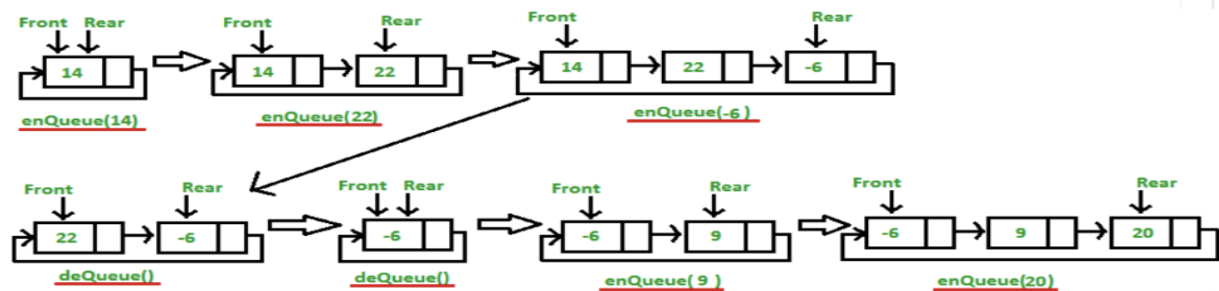
Question description

Lalitha is a B.Tech student. During her final year Campus Interview, she has an opportunity to get a job in a software company in Bangalore.

The company provides Five months training period with Rs.30000/month Package. Then it will be incremented to Rs.55000 per month.

At the end of the training, the examination was conducted for all freshers, Lalitha got a question paper and one of the questions comes under the concept of Queues in data structure that is Circular Linked List Implementation in Circular Queue.

Can you help?



Constraints

1. 10 elements inserted in array Q, next 9 elements inserted into array V, remaining elements inserted into array Z.

Input Format

First line indicates the number of elements to be inserted in the queues.

Second line indicates the elements.

Output Format

first line indicates the dequeue element of Q2

Second line indicates the dequeue element of Q1

Third line line indicates the dequeue element of Q0

Note: Refer sample input and output test cases

Logical Test Cases

Test Case 1

INPUT (STDIN)

10
1 2 3 4 5 6 7 8 9 10

EXPECTED OUTPUT

Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1
Deleted value = 2
Elements in Circular Queue are:3 4 5 6 7 8 9 10

Test Case 2

INPUT (STDIN)

10
16 17 18 19 20 11 12 13 14 15

EXPECTED OUTPUT

Elements in Circular Queue are:16 17 18 19 20 11 12 13 14 15
Deleted value = 16
Deleted value = 17
Elements in Circular Queue are:18 19 20 11 12 13 14 15

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *front = NULL;
struct node *rear = NULL;
```

```
void enqueue(int data)
{
    struct node *newnode;
    newnode = (struct node *)malloc(sizeof(struct node));

    newnode->data = data;
    newnode->next = NULL;

    if (front == NULL)
    {
        front = rear = newnode;
        rear->next = front;
    }
    else
    {
        newnode->next = front;
        rear->next = newnode;
        rear = newnode;
    }
}

void display()
{
    struct node *temp;

    if (front == NULL)
    {
        printf("Queue is empty\n");
        return;
    }

    temp = front;

    printf("Elements in Circular Queue are:");

    do
    {
        printf("%d ", temp->data);
        temp = temp->next;
    } while (temp != front);

    printf("\n");
}

void dequeue()
{
    struct node *temp;

    if (front == NULL)
    {
        printf("Queue is empty\n");
        return;
    }
}
```

```
temp = front;

if (front == rear)
{
front = rear = NULL;
}
else
{
front = front->next;
rear->next = front;
}

printf("Deleted value = %d", temp->data);
free(temp);
}

int main()
{
int n, data;

scanf("%d", &n);

for (int i = 0; i < n; i++)
{
scanf("%d", &data);
enqueue(data);
}

display();
dequeue();
dequeue();
display();
return 0;
}
```

Output:

Test_case 1

```
10
1 2 3 4 5 6 7 8 9 10
Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1Deleted value = 2Elements in Circular Queue are:3 4 5 6 7 8 9 10
```

Test_case 2

```
● aadityajaiswal@Aadityas-MacBook-Air: ~/Users/aadityajaiswal/Desktop/DSA Practise /stack.question
10
16 17 18 19 20 11 12 13 14 15
Elements in Circular Queue are:16 17 18 19 20 11 12 13 14 15
Deleted value = 16Deleted value = 17Elements in Circular Queue are:18 19 20 11 12 13 14 15
```